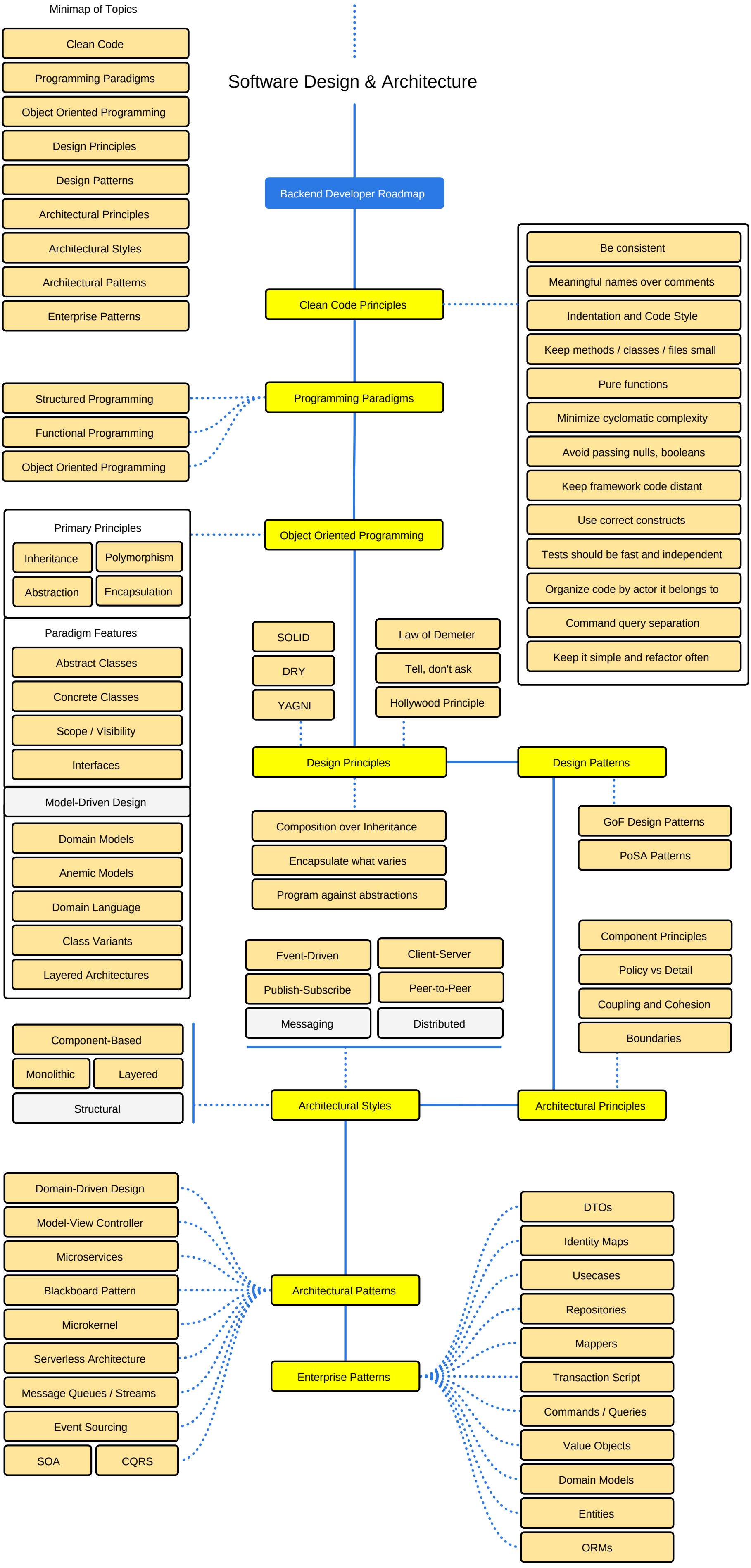
Minimap of Topics
Clean Code
Programming Paradigms
Object Oriented Programming
Design Principles
Design Patterns
Architectural Principles
Architectural Styles
Architectural Patterns
Enterprise Patterns
Software Design & Architecture
Backend Developer Roadmap
Clean Code Principles
Be consistent
Meaningful names over comments
Indentation and Code Style
Keep methods / classes / files small
Pure functions
Minimize cyclomatic complexity
Avoid passing nulls, booleans
Keep framework code distant
Use correct constructs
Tests should be fast and independent
Organize code by actor it belongs to
Command query separation
Keep it simple and refactor often
Structured Programming
Functional Programming
Object Oriented Programming
Programming Paradigms
Primary Principles
Inheritance
Polymorphism
Abstraction
Encapsulation
Object Oriented Programming
Paradigm Features
Abstract Classes
Concrete Classes
Scope / Visibility
Interfaces
SOLID
DRY
YAGNI
Law of Demeter
Tell, don't ask
Hollywood Principle
Design Principles
Design Patterns
Model-Driven Design
Domain Models
Anemic Models
Domain Language
Class Variants
Layered Architectures
Composition over Inheritance
Encapsulate what varies
Program against abstractions
GoF Design Patterns
PoSA Patterns
Event-Driven
Client-Server
Publish-Subscribe
Peer-to-Peer
Messaging
Distributed
Component Principles
Policy vs Detail
Coupling and Cohesion
Boundaries
Component-Based
Monolithic
Layered
Structural
Architectural Styles
Architectural Principles
Domain-Driven Design
Model-View Controller
Microservices
Blackboard Pattern
Microkernel
Serverless Architecture
Message Queues / Streams
Event Sourcing
SOA
CQRS
Architectural Patterns
Enterprise Patterns
DTOs
Identity Maps
Usecases
Repositories
Mappers
Transaction Script
Commands / Queries
Value Objects
Domain Models
Entities
ORMs